Strong-Lensing Time Delays in a Spatially Modulated Closed Cosmology: A Matched Galaxy Benchmark and Conditional Berger–Hopf Response

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Abstract

We construct a reproducible comparison between a conventional galaxy lens and the optical response available from the Berger–Hopf envelope equations, using a common incident-light preparation and physical source-screen convention. The frozen R1 closed cosmology supplies propagation; a separate galaxy encounter supplies the illustrative strong deflection. Three compatible Jacobi maps determine the lens equation and arrival-time functional. We retain source-coordinate, impact-parameter and measured-redshift endpoint changes in the response. The coupled cosmological provider evolves six physical anchor amplitudes for one selected normalized spatial harmonic. The established static spatial phase channel has zero mixed metric response when its spatial current vanishes; this restricted null does not establish absence of other geometric responses. Conditional responses are separated from action-normalized predictions and tested against independently re-solved images. Nuisance projection explicitly retains source, lens, mass-sheet, line-of-sight and cosmological freedoms. The illustrative isothermal benchmark has images at $+1.067883$ and -0.711922 arcsec and a relative delay of 17.094532 days. The six conditional state responses have rank three with nuisances fixed and rank zero after the declared unrestricted nuisance projection. A small specified scalar-state probe changes the delay by -0.01718 seconds; its amplitude is not an action-derived envelope calibration. The checkpoint identifies the local common-domain action projections needed for a galaxy-replacement prediction.

Keywords: strong gravitational lensing; time delays; closed cosmology; Jacobi maps; mass-sheet degeneracy; Berger–Hopf framework

1 Physical question and scope

In the Berger–Hopf interpretation, gravitational lensing is the optical response of an envelope event traversed by environmental radiation. The calculation relates prepared incoming light to its outgoing observables, conditional on the envelope state and surrounding geometry. Light is the probe of that state dependence. Traversal alone supplies no claim of encapsulation, de-encapsulation, reflection, absorption or a new photon law.

The quantitative question is whether a change in the encounter geometry or envelope state produces image positions and relative arrival times that survive the available source, galaxy and

environmental degeneracies. We retain two explicit realizations: a conventional galaxy-potential benchmark, and a response constructed from the existing Berger–Hopf metric and optical equations. The first executable comparison holds the galaxy encounter fixed and applies the available smooth geometric correction. An action-derived replacement for the local galaxy encounter remains a separate unresolved calculation. A restricted zero response and an exact degeneracy are valid outcomes of the declared comparison.

This focused draft develops the accepted contribution to "Strong Gravitational Lens Modeling," with submission intended by 25 February 2027. It uses the inherited frozen R1 background within its supported domain $0 \leq z \leq 2.1$. It does not admit the current muon preparation as a lensing calibration. No equation evaluated below requires restarting that preparation or an inclusive pair-production four-point calculation.

2 Matched optical experiment

The observer, lens and source carry orthonormal local tetrads. Two Sachs screen vectors are parallel transported along a common fiducial ray; transverse coordinates are physical lengths in those screens and angles are radians in the local observer or lens frame. We normalize the past-directed affine ray by the observer frequency magnitude $\omega_o = |u_o \cdot k_o| = 1$ and use positive $1 + z_a = \omega_a/\omega_o$. The photon wavelength is assumed much smaller than the metric and encounter scales. Null propagation, not an additional photon interaction law, determines the transfer.

The source emits the same narrow transient marker with the same emission phase, local spectrum, polarization and transverse source coordinate s in both realizations. Its small-amplitude stress is neglected as a test beam. The reference observed central wavelength is 550 nm; at the fixed source redshift the corresponding emission wavelength is 275 nm. The entrance and exit surfaces of the local encounter are planes in the lens rest frame, normal to the fiducial ray, outside the region assigned to the thin galaxy potential. Their thin-plane limit is the lens screen. Propagation outside this encounter uses the smooth reference geometry. The observable is the direction of each received image and the observer proper-time separation between copies of the same marker. Absolute unlensed travel time and a common source-dependent arrival constant are not measured.

The matching prescription fixes the transverse physical source coordinate, the galaxy's physical parameters and the measured lens/source redshifts $z_l = 0.3$ and $z_s = 1.0$. The reference redshift-selected screens consequently move when the metric response changes. At each perturbation images are solved again. Since

$$s = D_{os}\beta, \qquad x = D_{ol}\theta, \tag{1}$$

holding s fixed gives

$$\delta\beta = -D_{os}^{-1}\delta D_{os}\,\beta. \tag{2}$$

Fixing s, β and the image positions independently would describe inconsistent experiments. An alternative fixed inferred angular source is a different protocol and must be labeled accordingly. The longitudinal position and emission event follow each realization's fixed observed-redshift source surface. The physical matching holds the transverse displacement fixed; it does not additionally fix a complete three-dimensional source world-line across the varied geometries.

Cosmological radiation-fluid perturbations (δ_r, v_r) belong to the background state, not to this incoming optical pulse. Muon/reference-state covariances are a third object. Fixing the pulse does not fix its outgoing field or an induced interface response.

3 Coupled state and metric response

The frozen background parameters are

$$(\Omega_{m0}, \Omega_{r0}, \Omega_{k0}) = (0.31, 9 \times 10^{-5}, -0.018), \quad H_0 = 67.4 \text{ km s}^{-1}\text{Mpc}^{-1}, \tag{3}$$

with retained action parameters $\lambda = 1$, $q = 3$ and $V_0/(M_{\rm Pl}^2 H_0^2) = 2.3986073449708067$. The background is reused without new shooting, calibration or broad stability scans.

For a selected normalized $n = 2$ real spatial harmonic, the physical anchor at $z_i = 2.1$ is

$$Y_i = (\zeta_i, \Pi_{2i}, \delta_{mi}^{\rm ADM}, \delta_{ri}^{\rm ADM}, v_{mi}^{\rm ADM}, v_{ri}^{\rm ADM})^T. \tag{4}$$

The nine-dimensional spatial $n = 2$ eigenspace is not reduced to two physical amplitudes. Selecting one spatial profile leaves six initial amplitudes. The historical two-column seed sets the four ADM fluid coordinates to zero and selects a restricted slice.

For the observer-dipole profile on a reference sphere of radius R_H, one possible normalized choice is

$$Q_{\hat{p}}(\psi, \hat{n}) = \frac{\sqrt{3}}{\pi R_H^{3/2}} \sin(2\psi)\, \hat{n}\cdot\hat{p}, \quad \int_{S^3(R_H)} Q_{\hat{p}}^2 \mathrm{d}V = 1. \tag{5}$$

The dimensionless unit-sphere convention removes $R_H^{-3/2}$. The selected sky axis $\hat{p}$ is a conditional input, not a direction selected by an isotropic background or by the previous supernova fits. Its radius convention must agree with that of the action provider.

The action-native ADM plus Schutz–Sorkin implementation uses $x = (\zeta, D_m, D_r)$, with $D_m = \delta_m^{\rm ADM} + 3\zeta$ and $D_r = 3\delta_r^{\rm ADM}/4 + 3\zeta$. For auxiliary variables $u = (\alpha, \chi, v_m, v_r)$,

$$\frac{L_2}{a^3} = \frac{1}{2} u^T A u + u^T (B\dot{x} + Dx) + \frac{1}{2}\dot{x}^T K_0 \dot{x} + \frac{1}{2} x^T E x. \tag{6}$$

Constraint elimination gives

$$\begin{aligned} u &= -A^{-1}(B\dot{x} + Dx), & M &= K_0 - B^T A^{-1} B, \\ L &= -B^T A^{-1} D, & V &= E - D^T A^{-1} D, \\ p &= a^3(M\dot{x} + Lx), & \dot{x} &= M^{-1}(a^{-3}p - Lx), \\ \dot{p} &= a^3(L^T\dot{x} + Vx). \end{aligned} \tag{7}$$

These are the coupled original action applications evolved by the provider. The anchor definition $\Pi_2 = 2a_i^3 G_{S,2}(t_i)\dot{\zeta}_i$ is reference-normalized; it is generally not p_ζ. The ADM fluid continuity constraints set $\dot{D}_I = k_n^2(v_I - \chi)$ with the provider's closed-harmonic factor. Thus the reference normalization, reconstructed auxiliaries and Eq. (7) define the anchor-to-canonical map together. No independent momentum identification is imposed.

The reconstructed Newtonian rows use

$$\mathrm{d}s^2 = -(1 + 2\Phi)\mathrm{d}t^2 + a^2(1 - 2\Psi)\gamma_{ij}\mathrm{d}x^i\mathrm{d}x^j, \qquad \Phi_W = \frac{\Phi + \Psi}{2}. \tag{8}$$

Potential rows, harmonic normalization and time units are checked before composition. Legacy growth and luminosity-distance modules require a sign and provider adapter; their arrays are not substitutes for the coupled response at this state.

4 Three propagation maps and relative arrival times

For a null tangent and parallel screen, the Jacobi application is

$$\frac{\mathrm{d}^2 J_a}{\mathrm{d}\lambda^2} = \mathcal{T}(\lambda) J_a, \quad (J_a, J_a')|_{\lambda_a} = (0, \omega_a I), \quad \mathcal{T}_{AB} = -R_{\mu\nu\rho\sigma} e_A^\mu k^\nu k^\rho e_B^\sigma. \tag{9}$$

Then $D_{ab} = J_a(\lambda_b)$ maps angles in the frame at a to physical separations at b. The three maps D_{ol}, D_{os} and D_{ls} use one reference geometry and compatible frequency/screen normalization. Here J_a already includes the initial factor ω_a; it is not multiplied by that factor a second time. For reverse transport with the same oriented affine derivative, reciprocity reads $D_{ba}/\omega_b = -(D_{ab}/\omega_a)^T$. If the reverse angular opening is instead initialized toward the past endpoint, its derivative has the opposite sign and the positive physical reverse map obeys the corresponding plus-sign relation. The implemented reverse check uses that positive map. In closed FLRW they reduce to

$$\overline{D}_{ab} = a_b S_K(\chi_b - \chi_a) I, \qquad S_K(r) = \frac{\sin(\sqrt{K} r)}{\sqrt{K}}. \tag{10}$$

In particular D_{ls} is independently initialized at the lens, not the difference $D_{os} - D_{ol}$. The main encounter is omitted from all three reference maps when represented as a separate potential.

The existing seven-state reduction is recovered by writing the conformal Jacobi map as $S_K(I + \mathcal{A})$, with $\mathcal{A} = -\kappa I - \left(\begin{smallmatrix} \gamma_1 & \gamma_2 \\ \gamma_2 & -\gamma_1 \end{smallmatrix}\right)$. The directly evaluated sources give

$$Z' = \mathcal{Z}, \qquad \kappa'' + 2\frac{C_K}{S_K}\kappa' = \mathcal{F}, \qquad \gamma_A'' + 2\frac{C_K}{S_K}\gamma_A' = \mathcal{G}_A, \tag{11}$$

where $C_K = S_K'$, $\mathcal{F} = -\operatorname{tr}\delta\mathcal{T}/2$, $\mathcal{G}_1 = -(\delta\mathcal{T}_{11} - \delta\mathcal{T}_{22})/2$ and $\mathcal{G}_2 = -\delta\mathcal{T}_{12}$ in the rescaled conformal system. These fixed-coordinate path variables are passed through Eq. (17) before asserting a fixed measured-redshift response. They are not themselves a complete observed-redshift arrival kernel.

For the declared axial profile, the executed transport differentiates the original angular geodesic equation as an independent construction of the screen maps. Put $q(\chi) = Q_{\hat{p}}(\chi/R_H, \hat{p})$, $P_g = 1 + 2\Phi q$, $M_g = 1 - 2\Psi q$, $A = a\sqrt{P_g}$ and $B_g = a\sqrt{M_g}$. In conformal coordinates $\mathrm{d}s^2 = -A^2\mathrm{d}\eta^2 + B_g^2(\mathrm{d}\chi^2 + S_K^2 \mathrm{d}\Omega^2)$. With $N = \ln a$ and static-tetrad frequency E, the equations evaluated are

$$\begin{aligned}
N_{,\chi} &= -aH\sqrt{M_g/P_g}, \\
(\ln E)_{,\chi} &= aH\sqrt{M_g/P_g} - \frac{aH\Psi_{,N} q}{\sqrt{M_g P_g}} - \frac{\Phi q_{,\chi}}{P_g}, \qquad \lambda_{,\chi} = B_g/E, \\
\vartheta_{,\chi} &= \frac{L_\vartheta}{B_g S_K^2 E}, \qquad (L_\vartheta)_{,\chi} = E B_g a^2 q \left(\frac{\Phi}{A^2} + \frac{\Psi}{B_g^2}\right)\vartheta, \qquad D = B_g S_K \vartheta.
\end{aligned} \tag{12}$$

The matter tetrad follows the action-native velocity-potential convention $u_\chi = v_m q_{,\chi}$, rather than an unadapted legacy velocity row. For $s_u = u_\chi/B_g$ and $\gamma_u = \sqrt{1 + s_u^2}$, $\omega = E(\gamma_u + s_u)$. Observed-redshift endpoints are roots of $\omega = 1 + z$. Observer initialization includes the corresponding aberration $\vartheta_o = \gamma_{uo} + s_{uo}$; at the moving lens, $\vartheta_l = 0$ and $L_{\vartheta l} = B_{gl} S_{Kl} \omega_l$. Both screen maps coincide on this axis by rotation symmetry. In particular the angular forcing term in Eq. (12) is retained: suppressing it because the selected profile is a dipole would miss its focusing and endpoint response. Finite perturbations use this explicitly declared truncated metric only to check first derivatives; they are not nonlinear cosmological solutions.

A separate conformal Jacobi control applies the inherited seven-state equation to all six anchor columns, with axial shear zero and forcing

$$\mathcal{F} = q(\Psi'' - 8K\Psi) + 2\Psi'(q' + H_K q) + (\Psi - \Phi)\frac{q}{S_K^2}, \qquad H_K = C_K/S_K. \tag{13}$$

Primes here differentiate along the background χ ray. This is the stable harmonic form of $(\Psi q)'' + 2H_K(\Psi q)' - q(\Phi + \Psi)/S_K^2$; it retains the angular geodesic term. Its observer initialization includes the matter-tetrad aberration. The measured-redshift endpoint response is then $\delta \ln D_{ob} = \delta N_b + H_{Kb}\delta\chi_b - \kappa_b$, which is compared with the independently varied physical maps for $b = l, s$. The ζ_γ slot is unused and zero in this conformal control; the complete physical frequency and endpoint motion come from the metric ray of Eq. (12). A zero control slot therefore carries no assertion of a zero redshift response.

With $\hat{\alpha} = \nabla_x \hat{\psi}$ in the lens frame, the matrix lens and arrival-time equations are

$$\beta = \theta - D_{os}^{-1} D_{ls} \hat{\alpha}(D_{ol}\theta), \tag{14}$$

$$T = (1 + z_l) D_{ol}^T D_{ls}^{-1} D_{os}, \tag{15}$$

$$ct(\theta, \beta) = \frac{1}{2}(\theta - \beta)^T T(\theta - \beta) - (1 + z_l)\hat{\psi}(D_{ol}\theta) + f(\beta). \tag{16}$$

This is the physical angle-to-separation normalization of the arbitrary reference-spacetime formalism of Fleury et al. [1, Eqs. (2), (3), (11)–(14)]. Their frequency-weighted affine maps become the D_{ab} used here. Reciprocity implies symmetry of T when all maps come from the same reciprocal transport. The numerical antisymmetry residual is inspected before any arrival calculation. Arbitrary symmetrization would conceal an incompatible propagation input. Only differences $t_A - t_B$ are used, so $f(\beta)$ cancels.

4.1 Measured redshifts and moving initialization

The accumulated photon redshift response is $Z_a = \delta \ln(1 + z_a)|_{\lambda_a}$. Holding the measured redshift fixed moves the coordinate endpoint by

$$\delta\lambda_a = -\frac{Z_a}{(\ln \overline{\omega})'_a}, \qquad \delta\chi_a = -\frac{Z_a}{\mathcal{H}_a} \quad \text{in background conformal distance.} \tag{17}$$

Observer tetrad, source/lens velocity and gravitational frequency terms belong in Z_a for the chosen state. The map variation obeys

$$(\delta J_a)'' = \mathcal{T}_0 \delta J_a + \delta\mathcal{T}\, J_{a0}, \quad \delta J_a|_a = -\omega_a I \delta\lambda_a, \quad (\delta J_a)'|_a = \delta\omega_a I. \tag{18}$$

At its output one also adds $J'_{a0}(\lambda_b)\delta\lambda_b$ and the required screen/frame conversion. The zero of J_{a0} at its initial point removes the apparent $J''_{a0}\delta\lambda_a$ term there. Equation (18) is essential for D_{ls}; retaining only a fixed-affine Green integral does not give the measured-redshift observable. A frequency normalization expressed in physical angles must move with the initialization as well.

4.2 Complete first variation

Write $q = \theta - \beta$ and permit a physical encounter change $\delta\hat{\psi}|_x$. At a stationary image, varying Eq. (16) gives

$$\begin{aligned} c\delta t = {} & \frac{1}{2} q^T \delta T q - q^T T \delta\beta - \delta z_l \hat{\psi}(x) \\ & - (1+z_l)\left[\hat{\alpha}(x)^T \delta D_{ol}\theta + \delta\hat{\psi}|_x\right] + \delta f(\beta), \\ \delta T = {} & \frac{\delta z_l}{1+z_l} T + (1+z_l)[\delta D_{ol}^T D_{ls}^{-1} D_{os} + D_{ol}^T D_{ls}^{-1}\delta D_{os} \\ & - D_{ol}^T D_{ls}^{-1}\delta D_{ls} D_{ls}^{-1} D_{os}]. \end{aligned} \tag{19}$$

The image-position derivative cancels by stationarity; the source-coordinate and physical lens-impact terms do not. For the matched fixed-redshift protocol $\delta z_l = 0$, while endpoint shifts remain inside the map changes. If a local envelope potential has not been derived, its $\delta\hat{\psi}|_x$ is unresolved, rather than implicitly supplied by a smooth cosmological response. The executed correction holds the physical conventional encounter fixed and sets this term to zero by the definition of that restricted comparison.

5 Conventional encounter and analytic control

The numerical realization uses a singular isothermal sphere (SIS), with one-dimensional velocity dispersion $\sigma_v = 220\,\mathrm{km\,s^{-1}}$:

$$\hat{\psi}(x) = b_{\mathrm{phys}}|x|, \quad b_{\mathrm{phys}} = 4\pi\frac{\sigma_v^2}{c^2}, \quad \theta_E = b_{\mathrm{phys}}\frac{D_{ls}}{D_{os}}. \tag{20}$$

Here $\hat{\psi}$ has dimensions of length and b_{phys} is a dimensionless deflection amplitude. For scalar distances and $0 < \beta < \theta_E$, the two images and positive saddle-minus-minimum delay are

$$\theta_\pm = \beta \pm \theta_E, \qquad \Delta t_{-+} = \frac{2D_{\Delta t}}{c}\theta_E\beta, \quad D_{\Delta t} = (1+z_l)\frac{D_{ol}D_{os}}{D_{ls}}. \tag{21}$$

The source is away from the caustic and the central singularity. Its baseline position is $\beta_0 = (0.2\theta_{E0}, 0)$ and its held-fixed physical screen coordinate is $s = D_{os,0}\beta_0$. A smooth encounter control uses the same strength in $\hat{\psi}(x) = b_{\mathrm{phys}}\sqrt{|x|^2 + r_c^2}$ with $r_c = 0.05\,\mathrm{kpc} = 5\times10^{-5}\,\mathrm{Mpc}$. Its central image, when present, is retained in the image census rather than discarded to imitate the singular two-image case. The SIS is an illustrative conventional galaxy model, not an inferred mass distribution or a fit to observations. A local encounter specified by Eq. (20) is counted once, outside the reference Jacobi maps.

6 Envelope construction and the available channel

The inherited constitutive chain is

$$\begin{aligned} \mathcal{L}_\psi\psi &= J_{\mathrm{grav}} + \mathcal{E}_\Sigma[Q_\Sigma] + \xi, & h_{\mathrm{topo}} &= \mathcal{R}_{g\psi}\psi, \\ \mathcal{R}_{g\psi} &= -\mathcal{A}_g^{-1}\mathcal{B}_{g\psi}, & Q_\Sigma &= \frac{c^2}{8\pi}\int_{\Sigma_\star} H_\Sigma \Delta K_{uu}\mathrm{d}A. \end{aligned} \tag{22}$$

The inverse and projections require the same physical constraint domain, boundary conditions and state. In the retained static spatial limit, $\mathcal{L}_\psi$ reproduces $-\nabla^2 + B\nabla^4$. The static seam export has the inherited radial shape

$$\mathcal{E}^{(0)}_\Sigma[Q_\Sigma] = \epsilon_\Sigma \frac{Q_\Sigma}{Q} \frac{\Theta(r - r_\star)}{r^2}. \tag{23}$$

Its common-action normalization and orientation remain unresolved. The time derivative of this source for a moving seam is a proposed kinematic continuation; it is not used as an action-selected optical interaction in this checkpoint.

For normalized physical metric basis columns $P_g = (e_A, e_\psi)$ and a normalized local topographic profile φ, define

$$A^{\rm loc}_g = P_g^\dagger \widehat{H}_{gg} P_g = \begin{pmatrix} a & b \\ b & c \end{pmatrix}, \quad B^{\rm loc}_{g\psi} = P_g^\dagger \widehat{H}_{g\varphi}\varphi = \begin{pmatrix} u \\ v \end{pmatrix}. \tag{24}$$

The inverse application and local Schur curvature are

$$\begin{aligned} \mathcal{R}^{\rm loc}_{g\psi} &= \frac{1}{ac - b^2} \begin{pmatrix} bv - cu \\ bu - av \end{pmatrix}, & \Delta_g &= ac - b^2 \neq 0, \\ k_{\rm loc} &= c_\varphi - (B^{\rm loc}_{g\psi})^T (A^{\rm loc}_g)^{-1} B^{\rm loc}_{g\psi}, & c_\varphi &= \langle \varphi, \widehat{H}_{\varphi\varphi}\varphi \rangle. \end{aligned} \tag{25}$$

The diagnostic is $\|A^{\rm loc}_g \mathcal{R}^{\rm loc}_{g\psi} + B^{\rm loc}_{g\psi}\|$, with the determinant, basis norms, Legendre gate and boundary-domain status reported. These coefficients are action projections, not free fitting parameters. The cosmological lapse/shift auxiliary Hessian in Eq. (6) does not automatically supply this local physical metric/topographic Hessian.

The available same-domain action projection is nevertheless executed on the ADM auxiliary block at $z_l = 0.3$. After eliminating the fluid velocity auxiliaries its lapse/shift entries, in the action-density convention L_2/a^3 and unit normalized harmonic coordinates, are

$$(a_{\rm ADM}, b_{\rm ADM}, c_{\rm ADM}) = (-7.96725028,\ 0.56954480,\ 0.48267978), \qquad \det A_{\rm ADM} = -4.17001193. \tag{26}$$

All six mixed source columns are evaluated; their inverse response has residual 1.42×10^{-15}, the original action constraints have maximum residual 5.28×10^{-16} and the Schur reduction agrees with the native reduced action to 4.16×10^{-16}. The unit ζ source has pre-elimination self entry 0.301115412 and Schur entry 5.097094780. The reduced kinetic eigenvalues are $(0.00418211, 0.17251064, 0.43386354)$, so its Legendre gate is satisfied at this event. These data preserve the common cosmological domain and demonstrate its actual auxiliary inverse application. The map $\Phi = \alpha + \dot{\chi}$, $\Psi = -\zeta - H\chi$ depends on state and time derivatives, so it cannot rename this block as the local physical $A^{\rm loc}_g$ of Eq. (24).

In the established static block-diagonal spatial phase channel the mixed source is explicitly

$$\begin{aligned} \mathcal{S}_{ij}[j, \varphi] &= 2F''(X)(j^k \nabla_k \varphi) A_{ij} + 2F'(X) j_{(i} \nabla_{j)} \varphi - F'(X) g_{ij} (j^k \nabla_k \varphi), \\ (u, v) &= \int \mathrm{d}\mu_g \, w \, ((e_A)^{ij}, (e_\psi)^{ij}) \mathcal{S}_{ij}. \end{aligned} \tag{27}$$

For $j_i = 0$, or for $\nabla_i \varphi = 0$, the pointwise source and both projections vanish exactly. On any invertible admissible metric block this gives $\mathcal{R}^{\rm loc}_{g\psi} = 0$ and $k_{\rm loc} = c_\varphi$. This is a channel-specific null result; it does not determine a nonzero-current state or prove that all lensing vanishes. No new channel is introduced to manufacture a signal. For a constant texture admitted by the domain, the phase-invariant $F(X)$ sector also has zero self entry; other action and boundary terms must be included before identifying the complete c_φ.

The smooth coupled cosmological metric response is separately available from Eqs. (6)–(8). Its transfer per anchor amplitude is a conditional state sensitivity. Until a physical envelope state and the common-domain local coefficients are supplied, that sensitivity is neither a galaxy-replacement amplitude nor a calibrated BHSM prediction.

7 Nuisance projection and distinguishability

Let the local observable vector contain the image coordinates and independent relative delays. In the linearized model

$$y = BX + N\eta + \epsilon, \qquad WCW^T = I, \quad P = I - (WN)(WN)^+, \quad F = (WB)^T P(WB), \tag{28}$$

B contains conditional physical-state columns, while N contains source coordinates, galaxy parameters, mass-sheet scale, independent line-of-sight contributions and cosmological nuisance columns admitted by the declared protocol. The covariance C is a specified illustrative measurement covariance, not an estimated observational likelihood. Singular values of PWB report the surviving rank and parameter-space null directions. On a null direction there is no likelihood-derived finite uncertainty. The zero eigenvalue of a Fisher pseudoinverse must not be reported as a zero error bar. A prior must be added and identified before it can supply missing information.

For a reduced scalar lens the mass-sheet transformation is

$$\psi_\lambda(\theta) = \lambda\psi(\theta) + \frac{1-\lambda}{2}|\theta|^2, \quad \beta_\lambda = \lambda\beta, \quad \Delta\phi_\lambda = \lambda\Delta\phi. \tag{29}$$

It preserves image positions under the corresponding source rescaling. An independent unrestricted delay scale for each lens absorbs a common rescaling of all its delays. More sky directions alone do not remove that freedom. Delay ratios can remove a common scale only where multiple independent delays exist; a two-image SIS has just one independent delay. Image distortions or independent environmental/kinematic data require their own response and likelihood.

The dominant-lens treatment permits subcritical line-of-sight corrections beyond a single external convergence/shear [2]. In particular observer–lens, observer–source and lens–source segments enter differently; treating only the observer–source convergence can bias cosmographic inference [3]. Their physical contributions are already contained in the three reference maps here and are not added again as a second convergence potential. An independent nuisance column represents uncertainty in an included contribution, not another copy of its mean.

The familiar H_0^{-1} delay scaling follows when dimensionless expansion history, reduced angular galaxy parameters and angular source coordinates are fixed: every distance scales as H_0^{-1}. Holding the physical source coordinate fixed instead changes β, and therefore requires the full variation in Eq. (19). The two conventions are not interchangeable. In the scale-free SIS control, the fixed physical transverse source gives an exact first-order cancellation of the delay response to $\ln H_0$. The independently checked fixed angular-source convention instead gives $\partial\Delta t/\partial \ln H_0 = -\Delta t$.

A pure sky dipole has zero trace-free angular Hessian at Born order, $(\nabla_a\nabla_b Q)^{\mathrm{TF}} = 0$; its shear null does not remove its redshift or isotropic focusing response. Likewise, for m coherent global amplitudes and a fixed response M_O,

$$C_O^{\mathrm{topo}} = M_O C_q M_O^T, \qquad \operatorname{rank} C_O^{\mathrm{topo}} \le \operatorname{rank} C_q \le m. \tag{30}$$

This bound concerns the isolated coherent contribution. One local random field with independent increments can generate many nonparallel response columns and need not have rank one. No Gaussian factorization of a dressed state or substitution of a retarded kernel for a statistical state kernel is used.

8 Executed numerical checkpoint

The generated table below is tied to the input manifest and machine-readable calculation rather than to a manuscript-only amplitude choice. Conditional perturbations use the same source convention as the benchmark. Centered finite changes re-solve the image equation independently before comparing arrival-time differences with Eq. (19).

Quantity	Executed result
Lens/source redshift	0.3/1.0
D_{ol}, D_{os}, D_{ls} [Mpc]	945.162839, 1689.312240, 1076.994348
SIS image x positions [arcsec]	1.0678831, -0.7119220
SIS relative delay [days]	17.09453205
Free-nuisance surviving state rank	0
Fixed-nuisance control rank	3
Synthetic-prior-assisted rank	3
Static spatial phase (u, v)	$(0, 0)$ under the stated current gate

The physical background maps are

$$(D_{ol}, D_{os}, D_{ls}) = (945.162839, 1689.312240, 1076.994348)\,\mathrm{Mpc}. \tag{31}$$

The source-screen displacement is $(1.457663367, 0)$ kpc. The SIS images are $(1.0678830532, 0)$ and $(-0.7119220355, 0)$ arcsec, with relative delay 17.0945320516 days. Their original lens-equation residual is below 8.5×10^{-22} radians and their arrival stationarity residual is zero at reported floating precision. The analytic scalar SIS delay agrees within 1.1×10^{-14} days. Baseline segment reciprocity residuals are below 9×10^{-16}, and the assembled time-delay matrix is symmetric without modification.

In the anchor order of Eq. (4), the executed fractional relative-delay sensitivities are

$$\begin{aligned}\frac{1}{\Delta t}\frac{\partial \Delta t}{\partial Y_i} = (&-1.16307 \times 10^{-5},\ 2.83592 \times 10^{-3},\ 2.79299 \times 10^{-4},\\ &7.83676 \times 10^{-8},\ -2.69536 \times 10^{-3},\ 5.31244 \times 10^{-7}).\end{aligned} \tag{32}$$

Each entry is per unit of its declared physical anchor coordinate with the normalized spatial profile $N_Q = 0.02709350135$ in $H_0 = 1$ units. The six columns use independent coupled fluid as well as scalar initial conditions; they are not the older restricted two-column calculation.

For the explicitly specified probe $Y_i = (10^{-3}, 0, 0, 0, 0, 0)$, the images shift by $(+4.07393 \times 10^{-8}, -5.56173 \times 10^{-8})$ arcsec along the image axis. The independently solved relative delay changes by -1.98831×10^{-7} days, or -0.0171790 seconds. The first-order prediction is -1.98821×10^{-7} days. On that trial ray, $\max|\Phi| = \max|\Psi| \simeq 1.98 \times 10^{-6}$ and the largest relative metric perturbation is 3.96×10^{-6}. This is a numerical trial of a declared state amplitude, not a realized or calibrated physical envelope state. Its additional local static spatial phase response under the stated zero-current condition is exactly zero.

The complete relative-delay derivative for the unit ζ_i column separates into approximately zero from δT, -7.14118×10^{-4} days from source-coordinate remapping, $+5.15297 \times 10^{-4}$ days from physical galaxy impact and zero from the held-fixed intrinsic encounter. Their sum is -1.98821×10^{-4} days per unit ζ_i. Dropping the source or impact term would give a different response. Across all six columns, independently re-solved half-step image/delay derivatives agree with the complete response to a maximum covariance-whitened relative difference of 1.45×10^{-7}. The independent

conformal seven-state control, including the measured endpoint shifts above, reproduces all six observer–lens and observer–source map derivatives to a maximum absolute difference of $9.72 \times 10^{-11}\, c/H_0$ per anchor unit.

For the declared covariance, image-coordinate errors are 0.001 arcsec and the delay error is 0.1 days, with zero cross-covariances. With all nuisances fixed, the nonzero whitened state-response singular values are $(7.81204, 2.86044, 0.00402428)$: three combinations of the six amplitudes remain null even in this control. The unrestricted source, galaxy, mass-sheet, line-of-sight and physical-source H_0 nuisance matrix spans all five observable rows. The surviving state rank is zero, with largest projected singular value 1.88×10^{-15} against the declared 7.81×10^{-8} rank threshold. The whole six-dimensional state space is therefore unconstrained by this experiment under these nuisance freedoms. Delay-only projection also has rank zero. The synthetic nuisance prior control has rank three; it supplies independent assumed information, not an empirical constraint or detection.

The smooth-core check retains three images, at approximately $(1.067837, -0.002257, -0.711817)$ arcsec. Its central and outer relative delays are 30.259543 and 17.093193 days with respect to the positive outer image, giving ratio 0.56488603. The ratio's mass-sheet derivative vanishes, as required, while its source, strength and core derivatives remain nonzero. Allowing the declared smooth-galaxy and source nuisances still leaves state rank zero. Thus a mass-sheet-invariant ratio does not by itself identify the conditional state in this realization.

The first comparison is valid in geometric optics, linear response of the smooth geometry, small angles, and the thin, nonrelativistic conventional galaxy encounter. It excludes caustic crossings under the small finite variations used in the derivative check. An extended envelope would require integrating its derived metric through the entrance/exit surfaces before justifying any thin-plane reduction. This calculation does not supply that reduction by assumption.

9 Dependencies and next calculation

The current result separates four kinds of input. The frozen background, coupled action evolution, conventional encounter solution, compatible Jacobi/arrival equations and static phase-channel null are derived or directly evaluated within their stated domain. The incident pulse, selected spatial profile/axis, source coordinate, galaxy parameters and illustrative covariance are specified physical or measurement inputs. Unit responses to physical anchor amplitudes and any independently declared optical forcing remain conditional sensitivities. The local envelope's common-domain projected Hessian, nonzero spatial-current reference state, topographic boundary/source profile, normalized seam export and any stochastic source covariance remain unresolved.

The next local envelope calculation must evaluate Eq. (27) on a specified current state and profile, project the constrained Hessian in Eq. (24), and solve Eq. (22) with the corresponding boundary conditions. It can then provide $\delta\hat{\psi}|_x$ or an extended metric encounter for Eq. (19). Smooth cosmological propagation already included in D_{ab} must remain excluded from that local encounter. This is the specific dependency needed for testing an additional interface response or a galaxy-replacement interpretation.

The benchmark and conditional transfer therefore support a controlled state-dependence experiment: a 17.094532-day conventional benchmark, six computed state-response columns, a restricted static phase null and an explicit failure of identification under the admitted nuisance freedoms. The rank-zero result is the distinguishability conclusion for this illustrative configuration. An independently normalized envelope response requires the local action/state/source closure above and independent information that removes the relevant degeneracy directions.

The reproducible entry point is `python scripts/run_strong_lensing_comparison.py` from

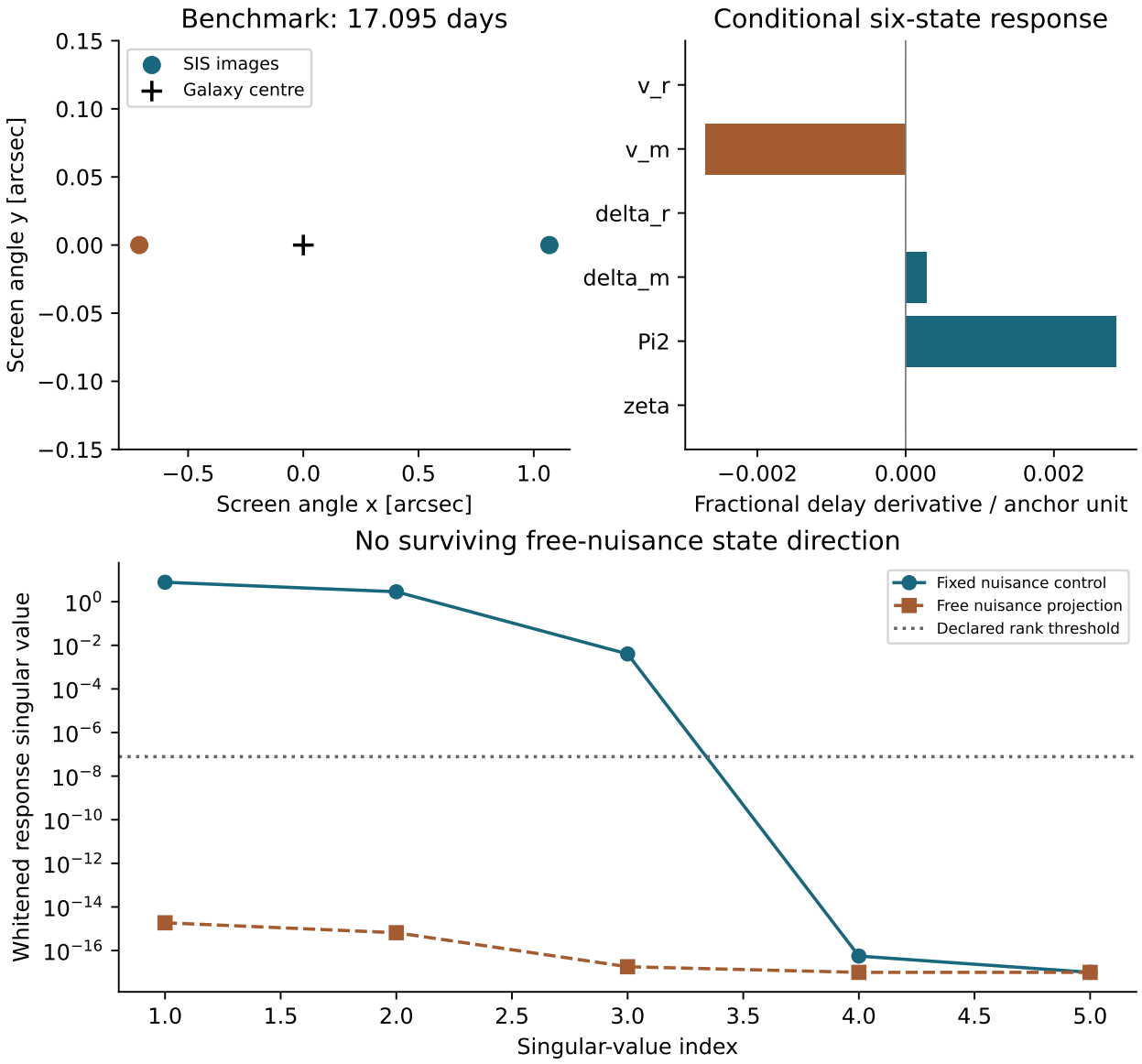


Figure 1: Executed illustrative comparison and conditional sensitivities. The input manifest defines the perturbation, covariance and source conventions. These are benchmark calculations rather than an observational fit or an action-normalized local envelope prediction.

the repository root. The machine-readable result, input hashes and response arrays are saved under `artifacts/strong_lensing_first_comparison_20261020/`. The focused tests cover the action projections, six-coordinate transport, redshift/reciprocity conventions, analytic lens control, complete arrival derivatives, mass-sheet transformation and correct treatment of unconstrained directions; 51 focused tests pass. The original broader manuscript remains a separate source.

References

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